

Design and modeling of curved beam based differential capacitive MEMS accelerometer

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Abstract—This work presents the design and modeling of micro-electro mechanical system (MEMS) accelerometer with curved beams as the suspension springs. Initially, we explore the mechanics of curved micro beams to find their natural frequency. Linear frequency analysis of curved beams have shown that as the radius of the curvature increases, the natural frequency decreases due to increase in the flexibility. Further, we utilized the curved micro beams as the suspension springs to design a MEMS accelerometer with differential capacitive sensing mechanism in MEMS+ software. The dominant in-plane mode of oscillation with natural frequency of 7.98 kHz has been considered for the modal, and electromechanical analysis. The obtained results have shown that the accelerometer has maximum mechanical, and capacitive sensitivity of $0.0308 \mu\text{m/g}$, and 27.211 fF/g respectively, for the radius of curvature of $2000 \mu\text{m}$.

Index Terms—MEMS accelerometer, Curved microbeam, Differential capacitance, Natural frequency.

I. INTRODUCTION

Micro-electromechanical systems (MEMS) accelerometer is one of most used sensors in various applications. A MEMS accelerometer is a sensor that measures force related to acceleration consisting of proof-mass of micron size suspended by beams, electronics and microfabrication technology into a single chip. When the device experiences acceleration, the proof mass moves with the device. The displacement is measured as an electrical signal in terms of capacitance or voltage. MEMS accelerometers have several advantages such as are small, lightweight, and cost-effective compared to traditional accelerometers. They are also compatible with integration into various electronic systems. However, they can be susceptible to external factors such as temperature changes, mechanical shocks, and electromagnetic interference. This can impact their accuracy and reliability. MEMS accelerometers are essential components in modern technology, enabling a wide range of applications that require accurate acceleration measurements.

A microelectromechanical system (MEMS) gravimeter with high sensitivity ($40 \frac{\mu\text{Gal}}{\sqrt{\text{Hz}}}$) and small size have been developed for measuring Earth tides [1] for gravity mapping applications such as land surveying via drones, volcanic monitoring, and multi-pixel density-contrast imaging arrays. An ultra-sensitive

seismic accelerometer employing geometric anti-spring technology is also designed to achieve nano-g sensitivity through an on-chip preloading system with curved leaf springs [2]. This mechanism enhances sensitivity by up to a factor of 26 and maintains linearity without power consumption. Thus, it showed low acceleration noise levels below $2 \text{ ng}/\sqrt{\text{Hz}}$ in a 50 Hz bandwidth using capacitive half-bridge read-out. Zhang et al., [3] explored the design of curved beam as suspension spring in a capacitive MEMS accelerometer. Liu et al., [4] examined the applications of high-performance MEMS sensors in various applications. Thus, the design of a MEMS accelerometer should involve analyzing various parameters to achieve optimal performance.

In the present study, we employ curved beam in a MEMS accelerometer design and study its influence on various performance characteristics using MEMS+.

II. MATHEMATICAL MODELING

A. Curved microbeam

According to the Inextensional theory of curved beams, it is assumed that the beam's centroidal axis and neutral axis coincide and it adds a constraint over extensional strain [$\epsilon = \frac{du}{ds} + \frac{w}{R} = 0$]. Hence, The governing differential equation of motion of curved beam configuration, in the transverse direction, based on Inextensional theory, can be written as [6],

$$EI \left[\frac{\partial^4 w(x, t)}{\partial x^4} + \frac{2}{R_c^2} \frac{\partial^2 w(x, t)}{\partial x^2} + \frac{w(x, t)}{R_c^4} \right] + m \frac{\partial^2 w(x, t)}{\partial t^2} = 0 \quad (1)$$

Here, E is Young's modulus, I is moment of inertia, $w(x, t)$ is transverse deflection, R_c is radius of curvature and m is mass of the beam.

The above equation (1), in terms of sector angle ($\theta = x/R$) can be written as,

$$\frac{EI}{R_c^4} \left(\frac{\partial^4 w}{\partial \theta^4} + 2 \frac{\partial^2 w}{\partial \theta^2} + w \right) + \rho A \frac{\partial^2 w}{\partial t^2} = 0. \quad (2)$$

Now, consider a set of boundary conditions for the curved microbeams, as illustrated in Figure 1. Employing the conventional analytical approach [7] for natural frequency calculation, we determined the natural frequencies for distinct curved beam configurations.

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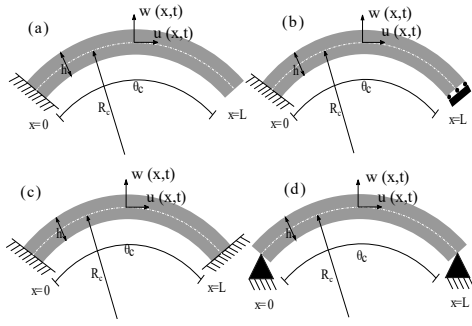


Fig. 1. Schematic of (a) Cantilever (b) Guided cantilever (c) Fixed-Fixed (d) Simply-Supported curved microbeam

- Curved cantilever beam

$$1 + \cos(p\theta_c) \cosh(q\theta_c) + \frac{1}{pq} \sin(p\theta_c) \sinh(q\theta_c) = 0 \quad (3)$$

- Guided curved cantilever beam

$$\sin(p\theta_c) \cosh(q\theta_c) + \frac{p}{q} \cos(p\theta_c) \sinh(q\theta_c) = 0 \quad (4)$$

- Fixed-Fixed curved beam

$$1 + \cos(p\theta_c) \cosh(q\theta_c) + \frac{1}{pq} \sin(p\theta_c) \sinh(q\theta_c) = 0 \quad (5)$$

- Simply-supported curved beam

$$\omega_n = \left(\frac{n^2 \pi^2}{L^2} - \frac{1}{R_c^2} \right) \sqrt{\frac{EI}{\rho A}} \quad (6)$$

where, $p = \sqrt{\frac{\omega_n}{\Omega} + 1}$, $q = \sqrt{\frac{\omega_n}{\Omega} - 1}$ are the frequency parameters and $\Omega = \sqrt{\frac{EI}{\rho A R_c^4}}$.

B. MEMS acclerometer

Let us consider the MEMS accelerometer as a spring-mass-damper system. From the force equilibrium, the governing equation of motion can be found as,

$$M\ddot{y} + C\dot{y} + Ky = Ma_e \quad (7)$$

where, M is the mass of the proof mass, C is the damping coefficient, K is the effective spring constant or stiffness, a_e external applied acceleration. Here, $\ddot{y}$, represents the time derivative of the displacement of the proof mass.

By taking Laplace transform on both sides of the equation (7), the transfer function of the spring-mass-damper system can be obtained as,

$$TF = \frac{y(s)}{a_e(s)} = \frac{1}{Ms^2 + Cs + K} = \frac{1}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (8)$$

where, ζ is called the damping ratio and ω_n angular natural frequency. By comparing the equation (7) and (8), we can write natural frequency f_n as ,

$$\omega_n = \sqrt{\frac{K}{M}} \rightarrow f_n = \frac{1}{2\pi} \sqrt{\frac{K}{M}} \quad (9)$$

Similarly,

$$2\zeta\omega_n = \frac{C}{M} \rightarrow \zeta = \frac{C}{2M\omega_n}. \quad (10)$$

Under the small damping assumption, the quality factor (Q) is defined as,

$$Q = \frac{1}{2\zeta} = \frac{M\omega_n}{C} \quad (11)$$

Further, under the static condition (all the time derivative terms goes zero in equation 7) of the MEMS accelerometer, from Hooke's law,

$$\frac{y}{a_e} = \frac{M}{K} = \frac{1}{\omega_n^2} \quad (12)$$

From the above equation it is evident that the sensitivity (output/input) is inversely proportional to the square of natural frequency of the system [8].

C. Capacitance modeling

Considering the differential capacitive sensing setup depicted in Fig. 2, when the accelerometer is in a stationary state (without external acceleration), the parallel plate capacitance can be determined using C_1 and C_2 . When an acceleration is exerted on the device, the proof mass shifts from its initial position, causing a change in the gap between the stator and rotor fingers. Hence, the C_1 and C_2 during the motion of the

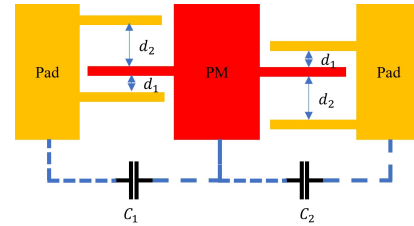


Fig. 2. Differential capacitance

accelerometer is defined as,

$$C_1 = \frac{\epsilon_r \epsilon_0 N_c A}{d_1 + y}, C_2 = \frac{\epsilon_r \epsilon_0 N_c A}{d_1 - y} \quad (13)$$

Therefore, the differential capacitance (ΔC) can be obtained as,

$$\Delta C = C_2 - C_1 = \frac{2\epsilon_r \epsilon_0 N_c A}{d_1} \left(\frac{y}{d_1} \right) \approx 2C_0 \left(\frac{y}{d_1} \right) \quad (14)$$

here, the term $C_0 = \frac{\epsilon_r \epsilon_0 N_c A}{d_1}$ is referred to as the nominal capacitance of the system. However, our modeling solely accounts for the impact of the gap d_1 on the overall capacitance, given that $d_2 \gg d_1$.

III. DESIGN AND MODELING OF MEMS ACCLEROEMTER

In this section we discuss the design and modeling methodologies of the proposed MEMS accelerometer. We utilized the MEMS+ software to simulate the fabrication process of silicon on insulator (SOI) technology for designing the accelerometer. The chosen material has Young's modulus (E), Poisson's ratio (ν), and density (ρ) as 167Gpa, 0.278, and 2329kg/m³, respectively. Following this, the fabrication process of the MEMS accelerometer, depicted in Fig. 3 (i), initiates from the

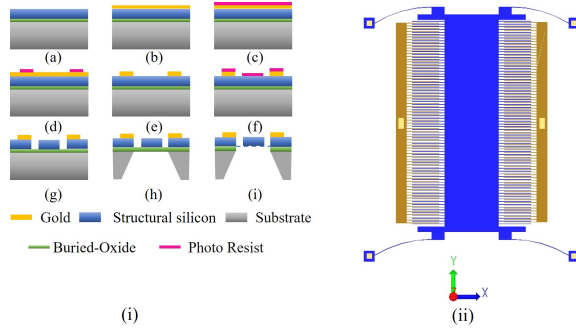


Fig. 3. (i) Schematic of fabrication flow of MEMS acclerometer (a) Silicon-on-Insulator (SOI) wafer, (b) Gold deposition for electrode, (c) Photoresist deposition, (d) Photoresist patterning, (e) Electrode patterning, (f) Photoresist patterning for MEMS structure, (g) Structure patterning, (h) Trenching from backside of wafer, and (i) Release of MEMS acclerometer by removing oxide material from backside. and (ii) Schematic of differential capacitive MEMS acclerometer with curved beam as suspension springs.

SOI wafer and progresses to the release of the free-standing structure.

Subsequently, We designed a MEMS acclerometer as shown in Fig. 3 (ii), with the parameters mentioned in Table I. In the design of the device we incorporated the curved beams as the suspension springs, which are be characterized by radius of curvature (R_c), thickness (t_s) and width (w_s), respectively. We also utilized the differntial capacitive sensing mechanism, which is characterized by finger spacing (d_1), anti-finger spacing (d_2), overlap length of the combs (l_o), respectively.

TABLE I
PARAMETERS OF MEMS ACCLEROMETER

Parameter	Value
Proofmass dimensions ($L_{PM} \times W_{PM}$)	$2000 \mu m \times 500 \mu m$
Thickness (t_s)	$25 \mu m$
Length of the comb (l_c)	$300 \mu m$
Width of the comb (w_c)	$6 \mu m$
Overlap length (l_o)	$250 \mu m$
Finger spacing (d_1)	$3 \mu m$
Anti finger spacing (d_2)	$10.5 \mu m$
Number of comb pairs (N_c)	77
Radius of curvature of curved beam (R_c)	$1000 \mu m$
width of curved beam (w_s)	$5 \mu m$

IV. RESULTS AND DISCUSSION

A. Frequency analysis of curved microbeams

In this section, we discuss the frequency analysis of curved microbeams for different boundary conditions with a radius of curvature of $500 \mu m$. From Fig. 4 it is noted that the natural frequency the curved microbeam decreases with the increase in sector angle from 5 deg to 90 deg for different boundary conditions. This is due reduction in the stiffness of the curved microbeam with increase in sector angle. Further, we compared our analytical results with numerical simulation in COMSOL multiphysics, and both have shown good agreement with each other.

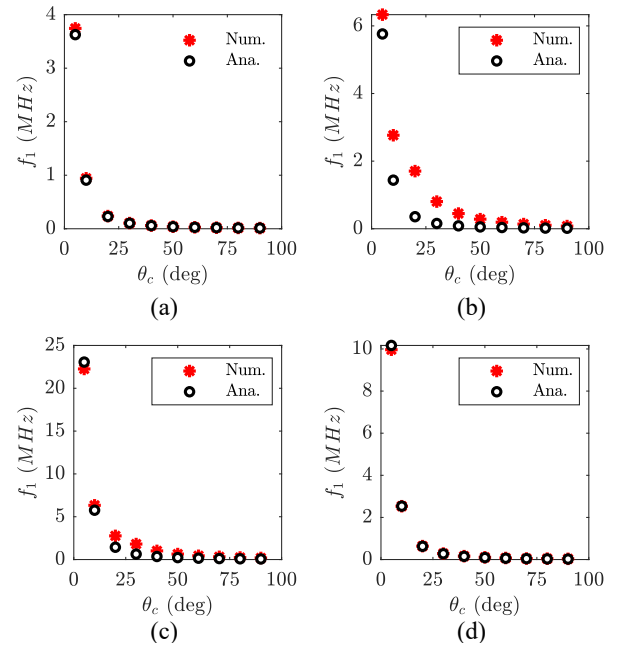


Fig. 4. First natural frequency of curved (a) cantilever, (b) guided-cantilever, (c) fixed-fixed, and (d) simply-supported microbeam.

B. Modal analysis

To begin with we performed the numerical modal analysis in MEMS+ software using Sparse modal solver to determine the resonance frequencies of the device. Fig.5(a) shows the

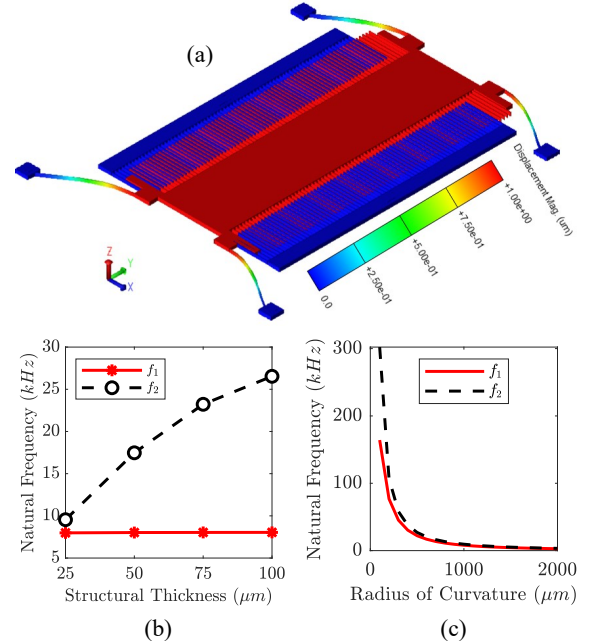


Fig. 5. (a) In-plane mode of MEMS accelerometer along y-axis, (b) Structural Vs. Natural frequency, and (c) Radius of curvature Vs. Natural frequency of MEMS accelerometer.

first in-plane mode along y-axis and with natural frequency of 7985.23 Hz . From the modes obtained it is observed that

the dominant mode is along y-axis with in-plane mode of oscillation. Additionally, we computed the analytical natural frequency of the in-plane mode using equation 5 and obtained a value of 8252.45 Hz, resulting in a maximum percentage error of 3.23. Hence, further analysis has been carried out based on the in-plane mode of vibration of the accelerometer.

We explored the relationship between the natural frequency (f_n) and the structural thickness (t_s). As depicted in Fig. 5(b), the in-plane mode natural frequency (f_1) remains unaffected by changes in structural thickness, while the out-of-plane mode natural frequency (f_2) rises with increasing structural thickness due to the increased flexural rigidity of the suspension springs. Likewise, we analyzed the impact of the radius of curvature (R_c) of suspension springs on the natural frequency. As illustrated in Fig. 5(c), it is evident that increasing the R_c leads to a decrease in the natural frequency (f_n), owing to the reduction in the effective stiffness of the curved suspension springs.

C. Electromechanical analysis

Whenever the device subjected to an external excitation, the proof mass tends to move from the original equilibrium position. The change in displacement of the proof is sensing using differential capacitive sensing mechanism. Fig.6(a) shows the maximum displacement experienced by the proof mass to the external excitation of $\pm 10g$. The results obtained clearly indicate that the maximum displacement of the proof mass increases with an increase in R_c , which can be attributed to the decreased effective stiffness of the curved suspension beams at higher values of R_c . Following that, we computed the change in capacitance (ΔC).

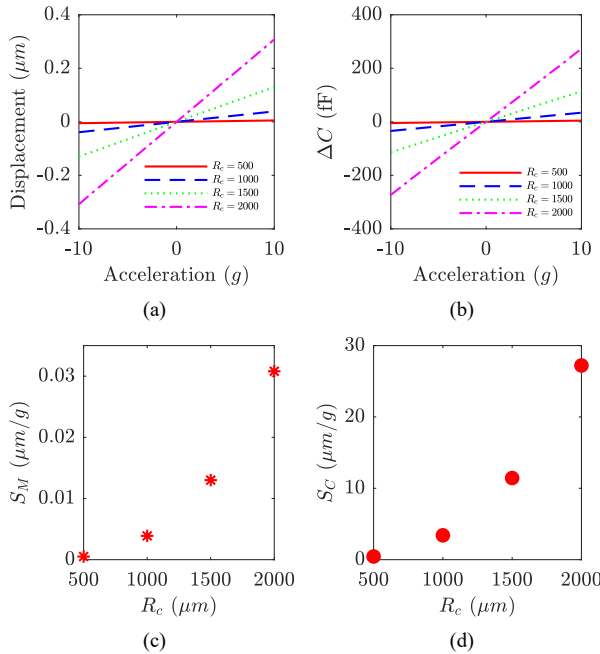


Fig. 6. Influence of structural radius of curvature (R_c) of curved suspension spring on the in-plane (f_1) and out-of-plane (f_2) modes of the MEMS accelerometer

As demonstrated in Figure 6(b), similar to the maximum displacement trend, it becomes apparent that with an increase in the radius of curvature, there is a corresponding increase in the change of capacitance. This is attributed to a reduction in the gap between the parallel plate comb pairs. Additionally, we computed the mechanical sensitivity (S_M = change in displacement/ input acceleration) and capacitive sensitivity (S_C = change in capacitance/ input acceleration). From the fig.6(c) and (d) the S_M and S_C of the MEMS accelerometer is found to be maximum with the values of 0.0308 $\mu m/g$, and 27.211 fF/g, respectively., for the radius of curvature 2000 μm .

V. CONCLUSIONS

This study outlined the designed and simulation of a micro-electromechanical system (MEMS) accelerometer utilizing curved beams as suspension springs. Initially, an exploration of the mechanics of these curved micro beams is undertaken to determine their inherent natural frequency. Linear frequency analysis reveals that increasing the curvature radius results in a decrease in the natural frequency due to enhanced flexibility. Subsequently, these curved micro beams are employed as suspension springs to devise a MEMS accelerometer incorporating a differential capacitive sensing mechanism using the MEMS+ software. The primary mode of in-plane oscillation, characterized by a natural frequency of 7.98 kHz. The results obtained indicate that the accelerometer achieves maximum mechanical and capacitive sensitivities of 0.0308 $\mu m/g$ and 27.211 fF/g, respectively, with a curvature radius of 2000 μm .

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